Chapter 9 - Analytic Geometry

#Review of Equations of Straight Lines

The equation $x = a$ is parallel to y-axis and $y = b$ is parallel to x-axis.

Standard Forms of Equation of Line

Slope-Intercept Form	$y = mx + c$	$m \rightarrow$ slope $c \rightarrow$ y-intercept
Double-Intercept Form	$\frac{x}{a} + \frac{y}{b} = 1$	$a \rightarrow$ x-intercept $b \rightarrow$ y-intercept
Normal/Perpendicular Form	$x \cos \alpha + y \sin \alpha = p$	$\alpha \rightarrow$ angle made by the perpendicular $p \rightarrow$ length of perpendicular

Point Forms of Equation of Line

Point Slope Form	$y - y_1 = m(x - x_1)$
Two Points Form	$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$

Linear Equation

An equation in the form of $ax + by + c = 0$ is said to be a linear equation.

1. Parallel equation: $ax + by + k = 0$
2. Perpendicular equation: $bx - ay + l = 0$

Reduction/Transformation of Linear Equation

Slope Intercept Form	$y = -\frac{A}{B}x - \frac{C}{B}$
Double Intercept Form	$\frac{x}{\frac{-C}{A}} + \frac{y}{\frac{-C}{B}} = 1$
Normal Form	$\frac{A}{\sqrt{A^2 + B^2}}x + \frac{B}{\sqrt{A^2 + B^2}}y = \frac{-C}{\sqrt{A^2 + B^2}}$

Point of Intersection of Two Straight Lines

Two straight lines $A_1x + B_1y + C_1 = 0$ and $A_2x + B_2y + C_2 = 0$ intersects at

$(x, y) = (\frac{B_1C_2 - B_2C_1}{A_1B_2 - A_2B_1}, \frac{C_1A_2 - C_2A_1}{A_1B_2 - A_2B_1})$ provided that $A_1B_2 - A_2B_1 \neq 0$.

Line through the intersection of two given lines

The equation of the line passing through the intersection of $A_1x + B_1y + C_1 = 0$ and $A_2x + B_2y + C_2 = 0$ is

$A_1x + B_1y + C_1 + k(A_2x + B_2y + C_2) = 0$ where k is any constant.

Condition of concurrency of three straight lines

The lines $A_1x + B_1y + C_1 = 0$, $A_2x + B_2y + C_2 = 0$ and $A_3x + B_3y + C_3 = 0$ intersect at a common point if

$$\begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix} = 0$$

Or, Some 3 numbers l, m, n exists such that

$l(A_1x + B_1y + C_1) + m(A_2x + B_2y + C_2) + n(A_3x + B_3y + C_3) = 0$

Angle between two straight lines

If θ is the angle between two straight lines $y = m_1x + c_1$ and $y = m_2x + c_2$ then $\tan\theta = \pm\frac{m_1 - m_2}{1 + m_1m_2}$

Condition of Parallelism: $m_1 = m_2$ **Condition of Perpendicularity**: $m_1m_2 = -1$

Important Points of Consurrencies

Orthocentre	The point where the perpendicular drawn from the vertices on their opposite sides meet.
Circumcentre	The point where the perpendicular bisectors of the sides of a triangle meet.
Incentre	The point where the bisectors of the internal angles of a triangle meet.
Centroid	The point where the medians of a triangle meet.

The two sides of straight lines

If $P(x_1, y_1)$ and $Q(x_2, y_2)$ are any two points and $Ax + By + C = 0$ is a straight line then

1. When $Ax_1 + By_1 + C$ and $Ax_2 + By_2 + C_2$ are of the same signs, P and Q lie on the same side.
2. When $Ax_1 + By_1 + C$ and $Ax_2 + By_2 + C_2$ are of different signs, P and Q lies on different sides.

#Length of the Perpendicular from a Point on a Straight Line po

Equation	**Length of perpendicular drawn from point** (x', y')
$x \cos\alpha + y \sin\alpha = p$	$\lvert x' \cos\alpha + y' \sin\alpha - p \rvert$
$Ax + By + C = 0$	$\frac{\lvert Ax' + By' + C \rvert}{\sqrt{A^2 + B^2}}$

#Bisectors of the Angles between Two Lines

The equation of the bisectors of the angles between two $A_1x + B_1y + C_1 = 0$ and $A_2x + B_2y + C_2 = 0$ is

$$\frac{A_1x + B_1y + C_1}{\sqrt{A_1^2 + B_1^2}} = \pm \frac{A_2x + B_2y + C_2}{\sqrt{A_2^2 + B_2^2}}$$